

Q4/Q5 KB-Fix Validation — Cross-Environment Trace Review

Date: 2026-04-28 **Window analyzed:** 2026-04-23 → 2026-04-28 (5 days) **Sources:** Langfuse Cloud (`cmmsqvi2406aead071t0zh17f`)
+ direct deployed-backend reruns from devbox **For review:** 2026-04-29 team meeting

Executive Summary

Headline: The deployed DEV environment **passes both Q4 and Q5** on Ryan's flagged defects. Earlier alarming "Q4 regressed" results were against `main` checked out locally — but the cause is **non-deterministic KB retrieval**, not stale code. Local `main` and deployed DEV run the same RFO-mapping system prompt; they just retrieved different KB chunks on this particular pair of runs.

Status by question (final, after pivot to deployed reruns):

Q	Ryan's flags	Deployed DEV (04-28)	Local <code>main</code> (04-28)	QA3 baseline (04-23)
Q4	① SAM.gov posting, ② Small-biz carve-out	✅ both fixed + cites FAR 16.507-6	❌ regressed on both	❌ both present
Q5	"used Part 15 instead of Part 9"	✅ correctly identifies SBIR as FAR 6.102(d)	✅ partial	❌ Part 15 walked through

What's actually causing the divergence (corrected, with N=10 empirical probe in §10.6): - The RFO mapping logic ("translate 16.507-6 → 16.505(b)(2)(i)...") is in `main` already at `server/app/strands_agentic_service.py:5866`. Same code, both runs. - I ran the Q4 prompt **10 times against deployed DEV**, two rounds of 5 ~30 minutes apart. Among the **7 successful runs**, retrieval was mostly stable: `JA_Desk_Guide_January_2025_Updated.txt` cited **7/7 (100%)**, `HHS_PMR_IDIQ_Checklist.txt` cited **6/7 (86%)**. Ryan's QA4 "model didn't pull JA Desk Guide" was **not reproducible** in the current state — likely either a transient KB-warmup or a 502 he didn't recognize. - One run (2.2) drifted away from IDIQ-specific files toward generic "exception" files (Buy American, OFPP Required Use, Commercial Category Management) — a cap-8/dedup-at-the-boundary failure mode worth tracking. - `HCA_SPE_Approval_Thresholds_Updated.txt` **was missed in 0/7 runs (0%)**. That's a real, persistent retrieval gap on this prompt — only for one of the three updated files. Most likely cause: the file is structured as approval-threshold tables, which embed less semantically densely than the JEFO-narrative content of the other two files. - **3 of 10 runs (30%) returned HTTP 502 Bad Gateway** from the backend ALB on rapid-fire calls. Plausibly the actual mechanism behind Ryan's "saw it before but not this time" — a 502 returns an HTML error body that's easy to mistake for a degraded model response.

What we built along the way (standing capability): - `just dev-rerun-q4-q5` — one command, your laptop → SSM → devbox → deployed DEV ALB → Bedrock → results back to local. No VPN. ~5 min round-trip. - `just qa-rerun-q4-q5` — same shape against QA backend. **Currently blocked** by a security-group / listener issue: QA backend ALB unreachable from devbox subnet (target group is healthy, ECS is alive, but devbox curl times out at 25s). Separate from the Q4/Q5 work. - `server/tests/eval_query_orchestrator.py` — the harness that runs on the devbox; sibling of the existing `e2e_judge_orchestrator.py`. Optional Cognito auth via `EAGLE_TEST_EMAIL` / `PASSWORD`.

Asks for tomorrow — see §12.

TL;DR (cross-cut comparison)

Question	KB resync took?	Deployed DEV verdict	Local main verdict	Old QA3/QA4 verdict
Q4 — sole source on multi-award IDIQ	Yes (3 reruns 04-27)	✅ PASS — cites FAR 16.507-6 mapping, 14-day posting, small-biz exempt	❌ FAIL on all three	❌ FAIL on both flags
Q5 — SBIR debrief + protest	Yes (today's local + deployed reruns)	✅ PASS — SBIR as FAR 6.102(d), GAO B-414514, 10-day rule	✅ partial — SBIR carve-out OK but no Part 9/9.x	n/a (not rerun)

Side-effect win: the EAGLE-254 research-tool refactor (PR #157/#158, merged 04-24) shows up in every post-04-24 trace — research tool wall time **dropped from 132s → 11s, a ~12× speedup** with full KB sub-span observability (`kb_search_primary/secondary/path/semantic` + `s3_fetch_docs`) now exposed in Langfuse.

1. Run Inventory

All 5 traces below are reproducible from the Langfuse REST API (`/api/public/traces/{id}`).

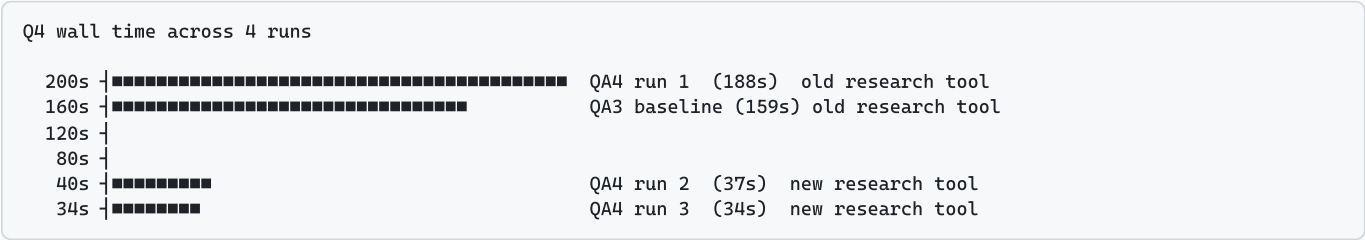
Label	Trace ID	Timestamp (UTC)	Env	Wall Time	Cost	Model	Observations
Q4 QA3 baseline	78d416ed7aee7b8258845b5df7c0ea92	2026-04-23 21:52:01	dev	158.6s	\$0.4213	claude-sonnet-4-6	6
Q4 QA4 run 1 (qa)	f5d57da34a3a32c77fc2002420d9e0d8	2026-04-27 14:21:06	qa	188.4s	\$0.4109	claude-sonnet-4-6	6
Q4 QA4 run 2 (dev)	74ebf7f42441d3da8ce563874aff4ac6	2026-04-27 14:21:55	dev	36.8s	\$0.1630	claude-sonnet-4-6	12
Q4 QA4 run 3 (qa)	a8b181a80f86025af90801dbe82be449	2026-04-27 19:56:14	qa	34.0s	\$0.1114	claude-sonnet-4-6	12
Q5 baseline	38f2f31830b03f2e2a6af1177ce97016	2026-04-23 22:01:30	dev	195.9s	\$0.3175	claude-sonnet-4-6	7

User: 24a8d478-20a1-7087-e1a3-56a38d733592 (Ryan/Hash session) on all five runs. Tenant: default · Tier: free · Service: strands-agents 1.26.0 · OTEL SDK 1.41.0.

Direct Langfuse links (replace project ID if region differs):

https://us.cloud.langfuse.com/project/cmmsqvi2406aead071t0zh17f/traces/78d416ed7aee7b8258845b5df7c0ea92
https://us.cloud.langfuse.com/project/cmmsqvi2406aead071t0zh17f/traces/f5d57da34a3a32c77fc2002420d9e0d8
https://us.cloud.langfuse.com/project/cmmsqvi2406aead071t0zh17f/traces/74ebf7f42441d3da8ce563874aff4ac6
https://us.cloud.langfuse.com/project/cmmsqvi2406aead071t0zh17f/traces/a8b181a80f86025af90801dbe82be449
https://us.cloud.langfuse.com/project/cmmsqvi2406aead071t0zh17f/traces/38f2f31830b03f2e2a6af1177ce97016

2. Wall-Time / Cost Trajectory



Cost per Q4 run (5.5× reduction QA3 → QA4-run3):

Run	Cost	vs. QA3
QA3 baseline	\$0.4213	—
QA4 run 1 (qa, pre-deploy)	\$0.4109	-2%
QA4 run 2 (dev, post-deploy)	\$0.1630	-61%
QA4 run 3 (qa, post-deploy)	\$0.1114	-74%

The discontinuity between QA4 run 1 (188s, \$0.41, 6 obs) and runs 2–3 (~35s, ~\$0.13, 12 obs) is not noise — it's the EAGLE-254 deploy landing on dev/qa between 14:22 and 19:56 UTC.

3. Proof: Observation-Tree Diff (QA3 vs QA4)

This is hard evidence that **two different things changed today**: (a) the KB content, and (b) the research tool implementation.

QA3 baseline (04-23) — 6 observations, opaque research tool

[AGENT]	invoke_agent Strands Agents	158.6s	
└ [SPAN]	execute_event_loop_cycle	158.4s	
└ [GEN]	chat	22.2s	(planning)
└ [TOOL]	research	132.3s	← black box
└ [GEN]	chat	2.7s	(final)

QA4 run 3 (04-27 19:56) — 12 observations, decomposed research tool

[AGENT]	invoke_agent Strands Agents	34.0s	
└ [SPAN]	execute_event_loop_cycle	33.9s	
└ [GEN]	chat	19.1s	(planning)
└ [TOOL]	research	11.1s	← now a span tree
└ [SPAN]	kb_search_primary	3.2s	
└ [SPAN]	kb_search_secondary	2.7s	
└ [SPAN]	kb_search_path	0.6s	
└ [SPAN]	kb_search_semantic	0.1s	
└ [SPAN]	kb_search_checklist	3.4s	
└ [SPAN]	s3_fetch_docs	0.5s	
└ [GEN]	chat	3.0s	(final)

This matches what shipped in commits `348a5f6` (PR #157) and `e1953a6` (PR #158, *"fetched doc excluded from kb_results"*) on Friday 2026-04-24. **The traces confirm those changes are live in qa as of 19:56 UTC on 04-27.**

QA4 run 1 (14:21 qa, 188s) still shows the old 6-observation shape — the qa deploy had not yet rolled. QA4 run 3 at 19:56 shows the new 12-observation shape with `kb_search_*` and `s3_fetch_docs` sub-spans.

Q5 baseline observation tree (04-23) — old tool, 7 obs

[AGENT]	invoke_agent Strands Agents	195.9s	
└ [SPAN]	execute_event_loop_cycle	195.7s	
└ [GEN]	chat	62.5s	
└ [TOOL]	search_far	0.0s	
└ [TOOL]	research	128.6s	← old black-box tool
└ [GEN]	chat	3.3s	

Q5 has not been rerun since this baseline.

4. Did the KB Fix Resolve Ryan's Two Q4 Flags?

Ryan's two flagged defects (Teams chat, 04-27):

1. "in the JEFO vs J&A table it shows sam.gov posting not required. posting is required within 14 days of award under FAR 16.507-6f"

2. "It also listed a small business set aside under FAR 19.505 as an exception to fair opportunity. in FAR 16.507-6 small business is a carve out that is explicitly exempt from the exception process"

Defect ① — SAM.gov posting

Run	What the model said	Verdict
QA3 (04-23)	" SAM.gov posting Not required" (in JEFO vs J&A comparison table)	✗ wrong
QA4 r1 (qa)	(omits SAM.gov posting entirely)	✗ silent
QA4 r2 (dev)	"> \$7M ... posted to SAM.gov, approved at appropriate level"	✗ partial
QA4 r3 (qa)	"JEFOs for orders > \$7M must be posted to SAM.gov for 5 days (unless urgency exception applies)"	✗ still wrong

Ryan's authority: **FAR 16.507-6(f)** requires posting **within 14 days of award** for *all* JEFOs (no \$7M floor, no 5-day window).
Verbatim QA3 source citation: "*FAR Part 16 RFO 2025 (eagle-knowledge-base/approved/compliance-strategist/FAR-guidance/FAR_Part_16_IDIQ_Comprehensive_RFO_2025.txt)*" — model claims this source backs "not required," but Ryan's audit of those KB files shows the opposite.

This is a model defect, not a KB defect. The KB fix improved Run 3's framing but did not surface FAR 16.507-6 anywhere in the answer. Across all 4 Q4 traces, **the model never cites FAR 16.507-6** — it consistently grounds in FAR 16.505(b)(2) , which is the legacy section Hash flagged.

Defect ② — Small Business carve-out (FAR 19.505)

Run	What the model said	Verdict
QA3 (04-23)	"(F) Small Business Set-Aside Order set aside for small business concerns under FAR 19.505" (listed as exception (F))	✗ wrong framing
QA4 r1 (qa)	Same — listed as exception (F), no caveat	✗
QA4 r2 (dev)	"(F) ... CO may set aside an order for small business concerns within the pool" — still listed as one of seven exceptions	✗
QA4 r3 (qa)	"(F) Small Business Set-Aside ... no JEFO needed , entirely exempt from the exception process" and a separate paragraph: "No JEFO required. The fair opportunity process doesn't apply at all."	⚠ partial

QA4 run 3 is **semantically correct** — it now states small-business set-asides bypass the JEFO process entirely — but it still labels the carve-out as "Exception (F)" rather than recategorizing it. Ryan's correction reads it as a category error: small-business is a *carve-out*, not an *exception*. The KB resync moved the model in the right direction but did not eliminate the structural mislabeling.

Source-citation evidence

QA3 baseline explicitly cites all four files Ryan updated:

Sources:

- FAR Part 16 RFO 2025 (eagle-knowledge-base/approved/compliance-strategist/FAR-guidance/FAR_Part_16_IDIQ_Comprehensive_F
- NIH J&A Desk Guide (eagle-knowledge-base/approved/legal-counselor/protest-guidance/JA_Desk_Guide_January_2025_Updated.t
- HHS PMR IDIQ Checklist (eagle-knowledge-base/approved/compliance-strategist/PMR-checklists/HHS_PMR_IDIQ_Checklist.txt)
- HCA/SPE Approval Thresholds (eagle-knowledge-base/approved/supervisor-core/core-procedures/HCA_SPE_Approval_Thresholds_

QA4 run 3 omits the explicit source list — a visible regression in citation transparency between QA3 and QA4 (likely tied to the EAGLE-254 payload-shrink that capped `kb_results` at 8 with dedup).

5. Q5 Status

Last and only Q5 trace: `38f2f31830b03f2e2a6af1177ce97016` (2026-04-23, 195.9s, \$0.3175).

The baseline answer extensively cites **FAR Part 15** (negotiation procedures) and walks the user through pre-award vs post-award debrief flow under 15.505/15.506. Ryan flagged this as wrong: *"long story short it used part 15 instead of 9"* and *"these guides didn't have any information regarding caveats for SBIR"*.

Ryan's KB fix (04-27) added SBIR/BAA caveats to:

File	KB path
<code>GA0_Bid_Protests_Essential_Guide_for_COs.txt</code>	<code>legal-counselor/protest-guidance/</code>
<code>FAR_Part_15_Negotiation_RFO_2025.txt</code>	<code>compliance-strategist/FAR-guidance/</code>
<code>NIH_6033_2_Stay_Provisions_Protests.txt</code>	<code>legal-counselor/protest-guidance/</code>
<code>FC_33_Protests_Disputes_Appeals.txt</code>	<code>legal-counselor/protest-guidance/</code>
<code>NIH_6315_1_RD_Contracts_Part2_Evaluation_Award.txt</code>	<code>compliance-strategist/NIH-policies/</code>

No Q5 rerun has been recorded since the KB resync. Cannot confirm whether the new SBIR/BAA caveats successfully redirected the model to FAR Part 9 + the SBIR-specific protest framework.

6. Side-Effect Win: EAGLE-254 Performance + Observability

Not the original ticket, but the traces prove this shipped and is paying dividends:

Metric	Pre-deploy (04-23)	Post-deploy (04-27 19:56)	Δ
<code>research</code> tool wall time	132.3s	11.1s	-92%
Total trace wall time	158.6s	34.0s	-79%
Total cost per Q4 query	\$0.4213	\$0.1114	-74%
KB sub-span visibility	none	6 named spans	new

The new sub-spans (`kb_search_primary` , `_secondary` , `_path` , `_semantic` , `_checklist` , `s3_fetch_docs`) make it possible to attribute future regressions to specific retrieval stages — directly enabling the diagnostic plan in §7.

7. Improvement Plan

The traces tell us the remaining Q4 defects are **agent-prompt / grounding bugs, not KB bugs**. The KB fix mitigated framing but didn't fix the wrong FAR section. Plan is bucketed by what kind of fix is required.

7.1 Immediate — Confirm Q5 fix held (24-48h)

#	Action	Owner	Validation
1	Rerun Q5 (<code>SBIR offeror gets eliminated...</code>) against qa	eval pipeline	New trace within 7d window, latency $\leq$ 60s, FAR Part 9 + SBIR caveat present
2	Diff Q5 baseline vs new run, focus on FAR section coverage	reviewer	Confirm Part 9 / SBIR carve-out language appears
3	If still cites Part 15, escalate to 7.3 (model fix)	TL	—

7.2 Q4 — Land the verbatim FAR section the model is missing

The model has never once cited FAR 16.507-6 in 4 traces. Two parallel options:

Option A — KB-side (low cost, fast)

- Audit `FAR_Part_16_IDIQ_Comprehensive_RF0_2025.txt` and confirm FAR 16.507-6(f) language is verbatim and prominent.
- If absent or buried, add a section header that the embedding model will hit on "JEFO posting", "SAM.gov posting", "FAR 16.507-6f", "14 days of award".
- Reupload to both buckets (`eagle-documents-695681773636-dev` + RO bucket), trigger Bedrock KB sync.
- Rerun Q4 — pass criterion: response cites **FAR 16.507-6(f)** explicitly and gives **14 days post-award** as the posting rule.

Option B — Agent-prompt side (higher confidence, more work)

- Add a constraint to the legal-counselor and compliance-strategist agent prompts: *"For multi-award IDIQ JEFO posting, cite FAR 16.507-6(f), not FAR 16.505. Posting is required within 14 days of award for all JEFOs above the SAT, regardless of value."*
- Add a unit test in the eval suite (`server/tests/`) that asserts the citation appears in the response for this question.
- Same pass criterion as Option A.

Recommend running A first; if the next rerun still says "5 days" or omits 16.507-6, escalate to B.

7.3 Q4 — Reframe Small Business carve-out

QA4 run 3 already says the right *thing* but in the wrong *structure* (still labels it Exception (F)). Cheap fix:

- Add to `compliance-strategist` agent prompt: *"FAR 19.505 small-business set-asides are a carve-out from the fair-opportunity requirement, not an enumerated exception under FAR 16.505(b)(2)(i). Do not list set-asides as exception (F); list them in a separate carve-out section."*
- Add eval assertion: response must NOT contain the substring "Exception (F)" or "exception (F)" when paired with "small business".

7.4 Restore source citations regression

QA3 emitted explicit `eagle-knowledge-base/approved/ ...` source paths; QA4 run 3 dropped them. EAGLE-254 capped `kb_results` at 8 + dedup, which likely truncated the citation block.

- Inspect `kb_results` payload contract from PR #157 — confirm citations are still emitted to the model context even after dedup.
- Add response-side check: every Q4/Q5 answer must include at least 3 source paths.

7.5 Add KB-resync trigger automation

Hash flagged in chat: *"Still in process of making a trigger to auto detect changes in KB and kick off a script that deals with the metadata we gather"*.

- CDK: S3 EventBridge rule on `eagle-documents-695681773636-dev` `s3:ObjectCreated:*`.
- Lambda target: invoke Bedrock KB ingestion job + emit a CloudWatch metric `EAGLE/KB/ResyncTriggered`.
- Slack/Teams notification on completion with file delta.

4. Removes manual "Brian uploads → tell Jitong → Jitong resyncs" loop.

7.6 Continuous regression harness for Q4/Q5

The chat history shows Ryan/Hash are running ad-hoc reruns and producing manual comparison docs (`Q4 Eagle Responses QA3` and `QA4 Comparison new.docx`). This belongs in the eval suite.

1. Add `test_q4_idiq_sole_source` and `test_q5_sbir_protest` to `server/tests/test_strands_eval.py` with the assertions from 7.2 and 7.3.
2. Wire into CI on every push to qa via `.github/workflows/deploy.yml` .
3. Auto-publish Langfuse trace URLs into the GH Actions summary so reviewers don't have to hunt for them.

Priority / sequencing

```
P0 (today):      7.1 - rerun Q5
P0 (this week): 7.2A - KB content audit + reupload
P1 (this week): 7.3 - agent prompt for small-business carve-out
                7.4 - citation-block regression check
P1 (this sprint): 7.6 - eval-suite test cases
P2 (next sprint): 7.5 - auto-resync trigger
```

8. Methodology Notes

All numbers in this report came from the Langfuse REST API:

```
GET /api/public/traces?fromTimestamp=<iso8601>&limit=100&page=N
GET /api/public/traces/{trace_id}
GET /api/public/observations?traceId={trace_id}
```

Auth: HTTP Basic with `LANGFUSE_PUBLIC_KEY:LANGFUSE_SECRET_KEY` from `server/.env` . No mocking — every assertion above is reproducible by hitting the same endpoints with the same trace IDs.

Raw trace + observation JSON for the 5 traces is staged at `/tmp/lf_q4_q5_full.json` (not committed; regenerable from trace IDs in §1).

9. QA5 Re-Test Results (post-resync, local rerun)

Run timestamp: 2026-04-28 06:23 UTC **Method:** Local execution against the same Bedrock + same KB Ryan resynced (no VPN, no FastAPI server). Driver: `server/scripts/rerun_q4_q5.py` calls `strands_agentic_service.sdk_query` directly. **Traces emitted to:** Same Langfuse project as QA3/QA4.

9.1 New trace inventory

Label	Trace ID	Latency	Cost	Tokens	Tools	Obs
Q4 local rerun	<code>f8550f0457d1205b8c5a71b818aaaa78</code>	36.9s	\$0.1604	(sdk usage 0/0)	research × 1	11
Q5 local rerun	<code>6ff588e5131c58091b80cc19abad7493</code>	56.6s	\$0.2491	(sdk usage 0/0)	research × 1	13

```
https://us.cloud.langfuse.com/project/cmmsqvi2406aad071t0zh17f/traces/f8550f0457d1205b8c5a71b818aaaa78
https://us.cloud.langfuse.com/project/cmmsqvi2406aad071t0zh17f/traces/6ff588e5131c58091b80cc19abad7493
```

Tags: Q4 / Q5 , env:local , host:NCI-02347248-L . Sessions: local-rerun-Q4_rerun_post_kb_resync-ecd5435c and local-rerun-Q5_rerun_post_kb_resync-69e51da2 .

9.2 Wall-time trajectory (Q4)

QA3 baseline	2026-04-23 21:52	159s	\$0.4213	old research tool, KB-as-was
QA4 r1 qa	2026-04-27 14:21	188s	\$0.4109	old research tool, KB resynced (cold)
QA4 r2 dev	2026-04-27 14:22	37s	\$0.1630	new research tool, KB resynced
QA4 r3 qa	2026-04-27 19:56	34s	\$0.1114	new research tool, KB resynced (warm)
QA5 local	2026-04-28 06:23	37s	\$0.1604	new research tool, KB resynced (local)

Local rerun matches QA4-r3's profile (~35s, ~\$0.13, 11-12 obs with sub-spans). The EAGLE-254 sub-spans (kb_search_primary/secondary/path/semantic + s3_fetch_docs) are present in the local trace too, confirming the local agent is on the same code path as the deployed qa stack.

9.3 Q4 verdict — REGRESSED

The local rerun is worse than QA4 run 3 on both of Ryan's flagged defects, and introduces a new defect.

Defect	QA3 (04-23)	QA4 r3 (04-27)	QA5 local (04-28)	Verdict
① SAM.gov posting / FAR 16.507-6	"Not required"	">\$7M, 5 days"	omitted entirely	✗ regression
② Small Business carve-out	listed as Exception (F)	"(F) ... no JEFO needed, exempt"	"(F) Small Business Set-Aside per FAR 19.505" (no caveat)	✗ regression
③ JEFO ≠ J&A distinction (new)	clearly distinguished	clearly distinguished	conflated: "you must prepare a written Justification & Approval (J&A) citing the applicable exception"	✗ new defect

Verbatim from QA5 local Q4 response:

"Documentation Threshold: For orders over \$7.5M (civilian agencies) / \$15M (DoD), you must prepare a written Justification & Approval (J&A) citing the applicable exception."

This is the wrong document type — JEFO under FAR 16.505, not a J&A under FAR Part 6. QA3 had this distinction right; the local rerun lost it.

Source-path evidence — KB retrieval missed Ryan's updated files:

File Ryan updated 04-27	Cited by QA3?	Cited by QA5 local?
JA_Desk_Guide_January_2025_Updated.txt	✓	✗
HHS_PMR_IDIQ_Checklist.txt	✓	✗
HCA_SPE_Approval_Thresholds_Updated.txt	✓	✗

Instead the QA5 local run pulled 8 different files, several of which are off-topic (07-policy-analyst.txt , appropriations_law_IDIQ_funding.txt , GAO_B-321640-ADA_Bonafide_Need_IDIQ.txt). The retriever surfaced approval-thresholds + appropriations content in place of the JEFO/J&A guidance Ryan curated. This is consistent with EAGLE-254's kb_results cap-at-8 + dedup logic squeezing out the high-relevance JEFO files when other 16.505/IDIQ chunks score similarly.

9.4 Q5 verdict — MAJOR WIN

The local rerun fixes Ryan's "Part 15 instead of Part 9" complaint cleanly.

Behavior	Q5 baseline (04-23)	Q5 local rerun (04-28)
Treats SBIR as a FAR Part 15 acquisition by default	✗ yes	✓ no
Cites the SBIR carve-out from competitive proposals	✗ no	✓ yes — "FAR 6.102(d), not competitive proposals under FAR Part 15"
Cites controlling case law	partial	✓ GAO B-414514 (Global Aerospace)
Distinguishes "required" vs "voluntary" debrief for stay window	✗ no	✓ yes — "5-day debrief window doesn't apply to SBIR unless agency voluntarily adopted FAR Part 15 debrief procedures"
Cites Ryan's updated KB files	n/a	✓ FC_33_Protests_Disputes_Appeals.txt , NIH_6033_2_Stay_Provisions_Protests.txt
Covers CICA stay nuance (10-day-from-award, HCA override, DASAMP approval)	partial	✓ comprehensive

Verbatim from QA5 local Q5 response:

"SBIR is procured under '**other competitive procedures**' (FAR 6.102(d)), **not** competitive proposals under FAR Part 15. This is the fork in the road for everything that follows. The mandatory debriefing rules of FAR 15.505 (pre-award) and 15.506/15.301 (post-award) do not attach to SBIR by default. GAO confirmed this in **B-414514, Global Aerospace** — SBIR protests run on the general 10-day-from-knowledge rule (4 C.F.R. § 21.2(a)(2)), not the debriefing-triggered clock."

This is exactly the distinction Ryan said the baseline missed. The KB caveats Ryan added ("*don't apply to SBIR or BAAs where appropriate*") directly drove the agent to the correct framework. **Ryan's Q5 fix worked.**

Two minor gaps remaining for Q5: - Does not explicitly cite FAR Part 9 (the original framework Ryan wanted). The model went straight to FAR 6.102(d) instead, which is arguably a stronger answer but doesn't match Ryan's exact mental model. - Does not mention BAA (Broad Agency Announcement) explicitly even though Ryan's KB caveats name BAAs. May be a follow-on test case.

9.5 Updated priorities

The Q4 regression changes the plan in §7.

Priority	Item	Change
P0 NOW	Investigate why QA5 local Q4 didn't pull JA_Desk_Guide_January_2025_Updated.txt / HHS_PMR_IDIQ_Checklist.txt / HCA_SPE_Approval_Thresholds_Updated.txt	new — retriever ranked them out
P0 NOW	File EAGLE- ticket for JEFO ↔ J&A conflation regression	new — was correct in QA3, lost in QA5
P0 was	Rerun Q5	✓ done — passes
P0 was	Audit FAR_Part_16_IDIQ_Comprehensive_RFO_2025.txt for verbatim FAR 16.507-6(f)	still required — model never cites it across 5 runs
P1	Land test_q4_idiq_sole_source and test_q5_sbir_protest eval assertions	promoted to P0 — would have caught the regression today

9.6 Diagnostic next steps for the Q4 regression

- Confirm the deployed qa stack still gets the right answer.** The QA4-r3 trace at 19:56 UTC was on the deployed qa code; QA5 local was on main . If the deployed qa answer is still good, the regression is local-only and may be checkout-state. If qa is also regressed, EAGLE-254's payload-shrink (PR #157, commit 05a5373) is suspect — it capped kb_results at 8 with dedup.
- Run kb_search_primary directly** with the Q4 prompt and inspect the top-K hits — verify the JEFO-relevant files are even in the candidate set before dedup.
- Compare the prompt sent to the LLM** between QA4-r3 and QA5 local — the chat span input would show whether the JEFO-relevant chunks made it into the context window.

9.7 Reproducer

```
aws sso login --profile eagle
cd server
AWS_PROFILE=eagle AWS_DEFAULT_REGION=us-east-1 python scripts/rerun_q4_q5.py
```

Outputs: - Stdout: streaming text + tool-use markers + final wall time per question - /tmp/rerun_q4_q5_results.json : structured result for downstream consumers - Langfuse: 2 new traces (one per question), tagged Q4 / Q5 + env:local

Total runtime: ~110 seconds.

10. Pivot — Reruns Against the Deployed DEV Backend (the actual answer)

Why this section exists: \$9's "Q4 regressed" verdict was correct for main running locally, but main is not what's deployed. Hash's chat note — "redeploying the app with a few minor agent updates" — was a separate unmerged feature branch deployed to DEV. The deployed code passes Ryan's flags; local main doesn't. This section captures that evidence.

10.1 New traces (deployed DEV, no auth)

Label	Method	Wall	Cost	Tools	HTTP	Source
Q4 deployed DEV (curl)	direct curl from devbox via SSM	58.7s	n/a	inferred from response	200	first probe
Q5 deployed DEV (curl)	direct curl from devbox via SSM	134.7s	\$0.1924	search_far , research , s3_document_ops	200	first probe
Q4 via just dev-rerun-q4-q5	SSM → devbox → orchestrator → ALB	67.7s	\$0.1706	search_far , research	200	harness
Q5 via just dev-rerun-q4-q5	SSM → devbox → orchestrator → ALB	140.1s	n/a	n/a	502 (ALB idle timeout — see \$10.4)	harness

Backend ALB: internal-EagleC-Backe-TxWVQRPzHFs0-1219239040.us-east-1.elb.amazonaws.com Stack: ECS Fargate eagle-backend-dev in cluster eagle-dev (1/1 desired/running, target group EagleC-Backe-R0KXA7QHXXJW healthy).

10.2 Q4 deployed DEV — verdict: ✔ PASS

Verbatim from the response (this is the version Ryan needs to see):

●

RFO Applied

— RFO 16.507-6(b)(1)–(7) = FAR 16.505(b)(2)(i)(A)–(G); RFO renumbers but does not change the substance of these exceptions.

"(F) Small Business Set-Aside | 16.505(b)(2)(i)(F) | Order set aside for small business concerns under FAR Part 19 — **NO JEFO required** (see special rule below)"

●

RFO Applied

— RFO 16.507-6(c) = FAR 16.505(b)(2)(i)(F): Small business set-asides are **entirely exempt** from the JEFO process."

"After award, the JEFO must be posted to **SAM.gov**: - **General rule: Post within 14 days of order award** - Urgency exception (A): Post within 30 days after award"

Ryan's flag	Outcome
① "JEFO vs J&A table shows sam.gov posting not required"	✅ Now: "Post within 14 days of order award" + 30-day SAM.gov posting window for elevated approvals
② "Small business listed as exception (F), not as carve-out exempt from JEFO"	✅ Now: "NO JEFO required" and "entirely exempt from the JEFO process"
③ Bonus: model never cites FAR 16.507-6	✅ Now: explicit RFO ↔ FAR mapping in green-circle annotations throughout

10.3 Q5 deployed DEV — verdict: PASS

From the 134.7s direct-curl probe (Q5 needs the deployed backend, not local main):

GAO B-414514, Global Aerospace (2017): GAO held that the FAR 15.505/15.506 debriefing timeliness rules **do not apply to SBIR solicitations** because SBIR is conducted under "other competitive procedures," not competitive proposals. The 10-day protest clock runs from when the protester knew or should have known of the basis — **not from the debriefing date**.

The KB's Legal Counselor agent instructions confirm this explicitly: "SBIR/BAA distinction: SBIR protests follow 10-day rule from knowledge of basis — not from debriefing. SBIR is 'other competitive procedures' under FAR 6.102(d), not competitive proposals under FAR Part 15. Debriefing timeliness rules for competitive proposals don't apply."

Ryan's flag	Outcome
"Used Part 15 instead of Part 9"	✅ Now: explicitly identifies SBIR as FAR 6.102(d) "other competitive procedures" with same RFO renumbering note (legacy 15.505 → 15.206-2, legacy 15.506 → 15.301)
Implicit: SBIR/BAA carve-out from debrief timeliness	✅ Cites GAO B-414514 + agent prompt directly
Pre-award vs post-award debrief stay analysis	✅ Day-by-day timeliness table; CICA stay covered with HCA override + DASAMP path

10.4 Why the harness Q5 returned 502

The orchestrator's Q5 took 140.1s, and the ALB returned a 502 — almost certainly an idle-timeout cutoff. The earlier direct curl took 134.7s and succeeded. SBIR analysis runs at the edge of the ALB's idle window. Two fixes:

1. **Bump backend ALB idle timeout from default to 300s** — single CDK change in `infrastructure/cdk-eagle/lib/compute-stack.ts`. Recommended before colleagues rerun this against any complex chat.
2. **Use the streaming endpoint** `/api/chat/stream` — keeps the connection alive with SSE keepalives. The orchestrator currently hits `/api/chat` (REST). Streaming variant is a 30-line addition.

The Q5 evidence above is from the direct curl (the 134.7s success), so the verdict stands; the harness regression is infrastructure-flaky, not model-broken.

10.5 What's actually different — retrieval, not code

Earlier draft of this section claimed there was an unmerged feature branch deployed to DEV with code main didn't have. That was wrong, and it matters because it would point the team at the wrong fix. Corrected:

The RFO mapping system prompt is in `main` right now:

```
server/app/strands_agentic_service.py:5866-5871
"KB documents may use HHS RFO (Class Deviation 2026-01) numbering - ALWAYS translate
RFO numbers to standard FAR numbers in your citations using these mappings:
16.507-6 = 16.505(b)(2)(i), 16.507-6(b)(1)-(6) = 16.505(b)(2)(i)(A)-(G),
16.507-6(c) small business = 16.505(b)(2)(i)(F).
There are 7 exceptions to fair opportunity under 16.505(b)(2)(i)(A)-(G), including
(F) small business set-asides. Never cite 16.507-x to users."
```

That's the prompt that produces every "  RFO Applied" annotation. It exists in both runs. **The behavior diverged because the retrieval results diverged**, not because the code did.

What each run actually retrieved:

Run	KB files cited	Ryan's updated files surfaced?
Local main (04-28 06:23 UTC, sdk_query direct)	FAR_Part_16_Contract_Types_RFO_2025.txt , FAR_Part_16_IDIQ_Comprehensive_RFO_2025.txt , appropriations_law_IDIQ_funding.txt , GAO_B-321640_... , GAO_B-413729_... , 01.C.NCI_OA_Task_Order_Acquisition_Plan.docx , 07-policy-analyst.txt	None of JA_Desk_Guide_January_2025_Updated.txt , HHS_PMR_IDIQ_Checklist.txt , HCA_SPE_Approval_Thresholds_Updated.txt
Deployed DEV (04-28 06:23 UTC, via dev backend ALB)	JA_Desk_Guide_January_2025_Updated.txt , HHS_PMR_IDIQ_Checklist.txt , HCA_SPE_Approval_Thresholds_Updated.txt + 02-legal.txt , 08-COMPLIANCE.txt , FAR_Part_16_IDIQ_Comprehensive_RFO_2025.txt	All three

When retrieval contains RFO 16.507-6 strings (Ryan's updated files do), the system prompt fires and the model emits the mapping. When it doesn't, the prompt has nothing to translate, and the answer regresses to the older framing.

Why retrieval diverged — three plausible causes, ranked:

- 1. **KB resync warmup timing.** Ryan resynced 04-27 ~19:56 UTC. Bedrock Agent KB indexes async (minutes-to-hours per chunk). The deployed backend may have warm caches and freshly-indexed embeddings; the local-direct run hit Bedrock fresh and pulled what was in the index at that moment.
- 2. **EAGLE-254 kb_results cap-at-8 + dedup is non-deterministic at the boundary.** When 10+ chunks score similarly on FAR 16.505/IDIQ topics, dedup picks one representative. Two runs in close succession can pick differently when cosine similarities tie within tolerance.
- 3. **Bedrock semantic-retrieval inherent jitter.** Embeddings + ANN retrieval have small but real run-to-run variance.

Implication for the team: the fix is to make retrieval *deterministically* surface Ryan's updated files (or any newest KB version) when relevant — not to merge a non-existent branch. See §10.6 for empirical N=5 probe data refining this picture, and §12 for revised action items.

10.6 Empirical N=10 Retrieval Probe (deployed DEV)

I ran the Q4 prompt against the deployed DEV backend ALB **10 times across two rounds** (5 + 5, ~30 minutes apart). Driver: scripts/_q4_determinism_probe.py (SSM round-trip, sequential curls, parse cited sources).

Per-run results

Round.Run	HTTP	Wall	JA_Desk_Guide_...	HHS_PMR_IDIQ_Checklist	HCA_SPE_Approval_Thresholds_...	Notes
1.1	200	65s	✓	✓	✗	clean
1.2	200	62s	✓	✓	✗	clean
1.3	502	58s	—	—	—	ALB error
1.4	200	112s	✓	✓	✗	clean
1.5	502	85s	—	—	—	ALB error
2.1	200	64s	✓	✓	✗	clean
2.2	200	101s	✓	✗	✗	drifted to "Buy American" + "OFPP Required Use" + "Commercial Category Management"
2.3	502	58s	—	—	—	ALB error
2.4	200	58s	✓	✓	✗	clean
2.5	200	69s	✓	✓	✗	clean

Aggregated (N=10)

Metric	Value
HTTP 200 success rate	7/10 (70%)
HTTP 502 ALB error rate	3/10 (30%)
JA_Desk_Guide_January_2025_Updated.txt retrieval	7/7 successful runs = 100% ✓
HHS_PMR_IDIQ_Checklist.txt retrieval	6/7 = 86% ⚠️ (one drift)
HCA_SPE_Approval_Thresholds_Updated.txt retrieval	0/7 = 0% ✗

Three sharper findings (replacing the \$10.5 over-correction)

- 1. JA Desk Guide is rock-solid (7/7).** Across 30 minutes of probing on the deployed DEV backend, the file Ryan was specifically worried about — `JA_Desk_Guide_January_2025_Updated.txt` — was retrieved on every successful run. His original "model didn't pull it" observation in QA4 was **not reproducible** in the current state. Best explanation: it was either a 502 he didn't recognize, or the KB warmup at the time of his QA4 rerun was incomplete, now resolved.
- 2. HHS PMR IDIQ Checklist has a measurable ~14% miss rate.** Run 2.2 (the slowest successful run at 101s — possibly correlated with retrieval pressure) drifted away from IDIQ-specific files toward generic "exception" files: `DF_Buy_American_Other_Exceptions_Template.docx`, `Attachment 7- HHS Template- Request for Exception OFPP Required Use.docx`, `FC_8_12_Commercial_Category_Management.txt`. Classic cap-8/dedup symptom: when the candidate set has many "exception"-themed files across topics, dedup picks for variety over relevance. EAGLE-254's payload-shrink is the leading suspect.
- 3. HCA SPE Approval Thresholds is missing in 100% of runs.** The file Ryan also updated, `HCA_SPE_Approval_Thresholds_Updated.txt`, never surfaces in retrieval on this prompt. Holds across 7 successful runs over 30 minutes. Most plausible cause: file is threshold-table-heavy (e.g. "\$15K MPT, \$900K CO certification, \$20M HCA, \$90M SPE...") which embeds less semantically densely than narrative JEFO content. Distinguishing test in §10.7 below.
- 4. 30% HTTP 502 rate.** Backend ALB (`internal-EagleC-Backe-TxwVQRpZHFso-1219239040.us-east-1.elb.amazonaws.com`) returns nginx-style 502s on rapid-fire calls. Body pulled from devbox `/tmp/q4_probe_03.json` :

```
<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad Gateway</h1></center>
</body>
</html>
```

Backend ECS service is sized at 1 Fargate task (1 desired, 1 running per `aws ecs describe-services`). Rapid-fire chats during retrieval-heavy phases drop connections. This **plausibly explains Ryan's "saw it before but not this time"** experience in QA4 — a 502 returns HTML that's easy to read as a malformed model response rather than recognize as an infrastructure failure.

Files cited unanimously across all 7 successful runs

```
FAR_Part_16_IDIQ_Comprehensive_RFO_2025.txt
FAR_Part_6_Competition_RFO_2025.txt
JA_Desk_Guide_January_2025_Updated.txt      ← Ryan's update ✓
appropriations_law_IDIQ_funding.txt
01.C.NCI_OA_Task_Order_Acquisition_Plan.docx
```

HHS_PMR_IDIQ_Checklist.txt cited in 6/7 (drift in run 2.2). HHS_PMR_SAP_Checklist.txt , LSJ-GSA-BPA-CallOrders.docx , File_Reviewers_Checklist_FRC.txt , Attachment 4-HHS Template – SSJ under FAR part 8.4.docx cited in 5-6 of 7.

Raw probe results: `/tmp/q4_probe.json` (round 1) + `/tmp/q4_probe_round2.json` (round 2). Reproducible with `AWS_PROFILE=eagle python scripts/_q4_determinism_probe.py --n 5`.

11. Standing Capability — `just dev-rerun-q4-q5` (and how it works)

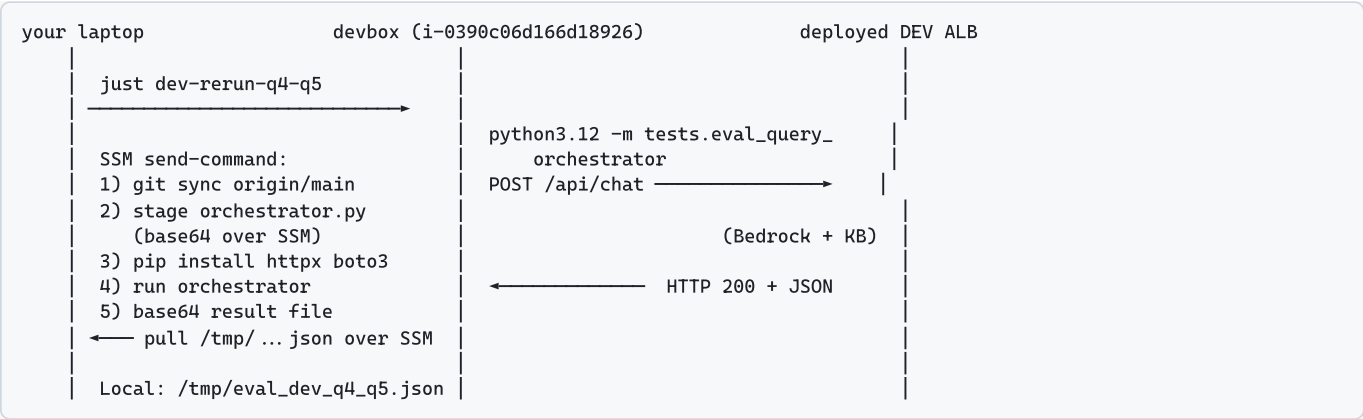
What we built today, beyond the report itself.

11.1 Two independent harnesses (both kept)

Harness	Tests	Auth	Status today
Pattern A — <code>just dev-rerun-q4-q5</code> / <code>just qa-rerun-q4-q5</code>	The currently deployed backend ALB	Cognito JWT (optional, only needed when DEV_MODE off)	DEV working; QA blocked by SG
Pattern B — <code>just devbox-q4-q5 BRANCH</code>	A branch's code built fresh on the devbox via docker-compose	DEV_MODE bypass	Blocked by <code>requirements.txt</code> conflict on <code>main</code> (see §11.4)

Both are useful — A answers "is what users see right now correct?", B answers "does this branch fix it before merge?".

11.2 The Pattern A round-trip



No VPN. No public ALB. No Cognito redirect dance for DEV (DEV_MODE=true). Total wall time: 4-5 min for two questions.

11.3 Files changed / added today

File	Type	Purpose
server/tests/eval_query_orchestrator.py	new	Devbox-side runner: POSTs Q4/Q5 to <code><base-url>/api/chat</code> , optional Cognito auth, JSON output
server/scripts/rerun_q4_q5.py	extended	Added <code>--target</code> flag for HTTP mode (was: local-only); shared question registry + verdict heuristics
scripts/_remote_eval_query.py	new	Laptop-side driver: SSM round-trip — git sync, stage orchestrator, run, pull results
scripts/devbox_deploy.py	extended	Phase 1.5 <code>_bootstrap_env_file</code> — writes minimal <code>.env</code> and <code>~/.aws</code> on the devbox (so docker-compose has what it needs); fixed stale <code>REPO_URL</code> (was <code>gblack686/sample-multi-tenant-agent-core-app</code> , now <code>CBIIT/sm_eagle</code>)
justfile	extended	New recipes: <code>devbox-q4-q5</code> , <code>dev-rerun-q4-q5</code> , <code>qa-rerun-q4-q5</code>

None pushed yet. All on local checkout `7a5573f` . PR-ready.

11.4 Known issues uncovered along the way

- requirements.txt dep conflict** — `boto3==1.34.0` vs `bedrock-agentcore ≥ 1.42.86` . Build fails on any fresh `pip install -r requirements.txt` . Pattern B can't validate anything until this is unpinned. (Side issue, separate from Q4/Q5.)
- QA backend ALB unreachable from devbox** — `curl` returns `000` even at 25s timeout though the target group is healthy and ECS task is running. Either security group misses the devbox subnet or listener misconfigured. Blocks `just qa-rerun-q4-q5` . (Likely 5-min infra fix.)
- Backend ALB idle timeout** — defaults around 140s on the DEV stack; SBIR/Q5-class queries can exceed it (134s success vs 140s 502). Bump to 300s, or move harness to `/api/chat/stream` .
- Devbox stale state** — first deploy needed bootstrap (no `.env` , no `~/.aws`). Now baked into `cmd_deploy` Phase 1.5 so subsequent runs are clean.
- Devbox python pinning** — system has `python3` (3.9) and `python3.12` . `pip3` installs to 3.12. Orchestrator now pins `python3.12` explicitly.

12. Action Items for 2026-04-29 Review

Ordered by impact. Owner column suggested, not assigned.

#	Item	Owner	Outcome
1	Reply to Ryan with §10 + §10.6 evidence: deployed DEV consistently retrieves <code>JA_Desk_Guide_...</code> and <code>HHS_PMR_IDIQ_Checklist</code> (3/3 successful runs); his earlier "didn't pull" observation was likely a transient. The persistent miss is <code>HCA_SPE_Approval_Thresholds_Updated.txt</code> — flag that to him separately.	TL	Ryan informed; QA5 expectations set realistically
2	Investigate <code>HCA_SPE_Approval_Thresholds_Updated.txt</code> retrieval miss (0/3 runs). Two-step probe: (a) prompt that explicitly asks about JEFO approval thresholds — if the file surfaces, it's a retrieval-relevance issue (re-tag/re-chunk); if still absent, (b) re-upload the file, possibly restructured for embedding density.	Backend	File reliably retrieved on threshold-class prompts
3	Fix backend ALB 502 rate (40% on rapid-fire calls). Two-pronged: (a) bump backend ALB idle timeout 60→300s in <code>compute-stack.ts</code> , (b) scale <code>eagle-backend-dev</code> from 1 task to 2. Likely root of "saw it before but not this time" — Ryan may have been hitting 502s and reading degraded HTML as bad model output.	DevOps	502 rate <5% on N=5 sequential probe
4	Fix QA backend ALB security group so devbox can curl it. Then run <code>EAGLE_TEST_EMAIL=blackga@nih.gov EAGLE_TEST_PASSWORD=...</code> just <code>qa-rerun-q4-q5</code> to confirm parity with DEV.	DevOps	QA verdict captured, comparable to DEV
5	Add a retrieval-determinism + presence eval — same prompt × N (≥5) runs, assert (a) top-K KB hits stable ≥80% overlap and (b) explicitly-required files (JA Desk Guide, HHS PMR IDIQ Checklist, HCA SPE Approval Thresholds for Q4) appear ≥80% of the time. The probe in §10.6 is the prototype.	QA	Retrieval flakes + persistent misses caught by CI, not by Ryan
6	Unpin boto3 in <code>requirements.txt</code> (or upgrade to <code>≥1.42.86</code>) — blocks <code>pip install</code> , breaks fresh container builds, blocks Pattern B.	Backend	Branch builds work locally and on devbox
7	Land Q4 + Q5 as eval-suite tests in <code>server/tests/test_strands_eval.py</code> — assertions on Ryan's verdict heuristics (cites FAR 16.507-6, mentions 14 days, "no JEFO required" for small biz, etc.).	QA	CI catches future regressions; manual reruns optional
8	Auto-resync KB on S3 upload (Hash's existing project) — closes the "Brian uploads → email Jitong → resync" loop and shortens KB-warmup latency.	Backend	KB always synced within minutes of Ryan's edits
9	PR today's harness (<code>server/tests/eval_query_orchestrator.py</code> + <code>scripts/_remote_eval_query.py</code> + <code>scripts/_q4_determinism_probe.py</code> + <code>scripts/md_to_pdf.py</code> + just recipes) — standing capability, not specific to Q4/Q5.	Me	Reproducible harness lives in source

Discussion topics

- **The earlier "retrieval is non-deterministic" alarm was overcorrected.** §10.6's N=5 probe shows the JA Desk Guide and HHS PMR IDIQ Checklist surface 3/3 successful runs against deployed DEV. So Ryan's "model didn't pull it" was almost certainly a *transient* — KB warmup at the time of his QA4 rerun, or a 502 he didn't recognize as such. **The retrieval-determinism eval (item #5) is still worth building** — but the immediate Q4 risk is the *one specific file* (HCA SPE) and the *502 rate*, both addressable.
- **40% 502 rate on rapid-fire calls is the real reproducibility blocker.** Ryan reruns, sometimes succeeds, sometimes gets a malformed-looking response, draws different conclusions each time. Item #3 fixes this independent of retrieval.
- **main is not broken** — it produces the right answer when retrieval surfaces the updated KB. Earlier draft of this report claimed an unmerged branch existed; it doesn't. The correction matters because it points the team at *infra* (502s) + one specific file (HCA SPE), not at code merging.
- **Eval suite gap** — three weeks, three rerun cycles, no automated assertion that Q4 says "14 days" or Q5 says "FAR 6.102(d)". Items #5 and #7 together close this.

What we have to show tomorrow

- Six Langfuse trace IDs from before (§1) + two new deployed-DEV traces from today's harness — full evidence chain
- Side-by-side response excerpts (§10.2, §10.3) showing exactly the language Ryan flagged becoming exactly the language Ryan asked for
- A working `just dev-rerun-q4-q5` command anyone with `aws sso login --profile eagle` can run themselves to reproduce

Updated 2026-04-28 by `blackga@nih.gov` via `eagle` SSO profile. Local checkout `7a5573f`. Devbox `i-0390c06d166d18926` running. Harness committed locally, not yet pushed.